What is Kubernetes Scheduler?

Professional Answer

The Scheduler assigns pods to nodes based on resource availability, constraints, affinity rules, and policies.

Unprofessional Answer

Scheduler decides where pods run.

What is the role of kube-scheduler?

The kube-scheduler is responsible for assigning pods to appropriate worker nodes.
It watches for newly created pods that do not have a node assigned.
The scheduler evaluates nodes based on resource availability and scheduling constraints.
It considers factors like CPU, memory, taints, tolerations, and affinity rules.
After selection, it updates the pod specification with the chosen node name.

What factors does the scheduler consider?

Professional Answer

- CPU & memory requests
- Node taints & tolerations
- Pod affinity/anti-affinity
- Node selectors
- Topology constraints

Unprofessional Answer

Resources and rules.

kube-scheduler (Scheduling Internals)

What it does internally

Scheduler works in **two main phases**:

Phase 1: Filtering (Predicates)

Eliminates nodes that cannot run the pod:

- Insufficient CPU/memory
- Taints not tolerated
- NodeSelector mismatch

- Volume conflicts

Phase 2: Scoring (Priorities)

Scores remaining nodes:

- Least loaded node
- Pod affinity
- Topology spread
- Custom scoring plugins

Final Step

- Highest score node selected
- Writes node name to PodSpec via API server

What is kube-scheduler?

Professional Answer

kube-scheduler is a control plane component responsible for selecting a suitable worker node for a Pod.
It watches for Pods that do not have a node assigned.
It evaluates nodes based on resources and constraints.
It assigns the Pod to the best-fit node.
It does not run Pods itself.

Weak Answer

Scheduler decides where pods go.

When does kube-scheduler act?

Professional Answer

kube-scheduler acts only when a Pod is created without a node assignment.
It does not manage running Pods.
It does not reschedule Pods automatically.
Once a Pod is bound to a node, the scheduler is done.
Re-scheduling happens only after Pod recreation.